

Electricity market design and renewable energy auctions: The case of Brazil

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ABSTRACT

Auctions have recently become the most widely adopted approach globally to procure renewable energy projects. Countries starting to use auctions often struggle to secure sufficient bidder interest – resulting in limited competition levels and high project prices. Brazil, a pioneer in the use of auctions to stimulate investment in renewable energy projects, offers important lessons on not only how auctions can be designed and implemented to stimulate competition and deliver effective investment outcomes, but importantly also shows how auctions form part of the country's electricity markets and can be integrated in a country's broader institutional and policy framework to ensure long-term success. This paper presents the detailing and specifications throughout the process of contracts and auction design definition to integrate renewable energy sources (RES) into the electrical matrix, from the institutional and procedural side to policy implications of securing the revenue stream and addressing off-taker risk between the sellers and buyers of electricity sector.

1. Introduction

Since the late 1990s, many countries had introduced competition into generation and energy trading activities to attract investments, ensure resource adequacy, increase access to electricity services, and to development of renewable energy sources (RES). Since then, auctions are spreading rapidly globally to procure RE. Brazil was pioneer and has one of the longest-running auction programmes. It offers important lessons on RE auction design and implementation. The development of RES in Brazil was the outcome of public policies and regulatory reforms implemented during the 1990s and 2000s. The Brazilian government launched the first institutional reform in 1995 to restore investment capacity and attract private capital to the power sector, after severe hyperinflation and fiscal crises in the 1980s.

The institutional model designed in 1995, however, was not able to guarantee security of supply, and Brazilian consumers had to endure rationing of energy in 2001. Consequently, on 15 March 2004, the federal government approved Act 10848 with the main aim of guaranteeing security of energy supply and resource adequacy in investment.

Regarding security of supply and the fairness of tariffs, the second reform promoted the use of energy auctions as the primary mechanism to procure energy and capacity with a long-term focus. These auctions

were very crucial for the beginning of the RES development in Brazil.

Between December 2004 and October 2019, the Brazilian programme contracted 9571 TWh of energy in 82 rounds of auctions, adding 105 GW of installed capacity which 77 GW were from RES. The success of the Brazilian auction programme depended to a large extent on the attention given by the public authorities to the design of the auctions, including the regulatory framework and the implementation process. In fact, the auctions have followed the same rules and structure since 2004, with only minor adjustments in contractual clauses and in the bidding mechanism.

Thus, Brazil has one of the longest-running auction programmes and it offers important lessons on RES auction design and implementation. Brazil's experience highlights the importance of considering three main goals when designing auctions. First, auctions must be attractive enough to investors to generate competition and to achieve optimal prices. Second, the auction design must ensure the commitment and reliability of the bidders and their technical and financial capability to build projects on schedule and deliver the promised energy. Third, the auction design should ensure the acquisition of the right mix of energy sources to safeguard the security of the electric system (Viscidi and Yépez, 2020).

To improve the attractiveness of the auctions, public authorities provided comprehensive information about the auction programme,

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including a schedule and technology allocations through a 10-year power system expansion plan, including generation and transmission.

To ensure timely project realisation, the bidding stage of the auction was preceded by a qualification phase to ensure the commitment and reliability of the bidders. Developers had to submit proof of land use rights and preliminary environmental permits and demonstrate technical and financial capability to build projects on schedule and deliver the promised energy. Bidders also needed to provide a bid bond before joining the bidding stage of the auction, and winners had to provide surety and performance bonds before contract signing.

To ensure energy security and the optimal mix of sources, the Brazilian auctions procured energy using both technology-neutral and technology-specific auctions. Reserve capacity was also procured by public auctions, using the same scheme and methodology developed for energy procurement.

Section 2 provides a theoretical background and literature review. Section 3 provides a brief introduction to the Brazilian power sector and describes the main policies to improve the integration of renewable energy sources into grid. In Section 4 we provide an overview of the regulatory and policy framework, including a description and analysis of the Brazilian auction design, including auction volumes, qualification criteria and processes, bidder ranking and winner selection and approaches to project derisking and credit enhancement. In Section 5 we present the main results. In Section 6 we present a reflection on the key lessons learned and recommendations. Finally, in Section 7 we provide some policy recommendations and conclude.

2. Theoretical background and literature review

The analysis of auctions as games of incomplete information originates in the work of William Vickrey (1961). Auctions are emphasized as an application of many extensions and applications and a powerful form of market-based allocation mechanisms, which provide a support cost efficient approach whenever there is a business environment with a high level of information asymmetry (McAfee and McMillan, 1987). Auctions are also important price disclosure instruments, and they can be adapted to diverse market designs in different environments and to target a particular technology or to be technologically neutral.

Auction performance in RES has been evaluated in the literature in different environments. The design of an auction can change both the competition and the attractiveness of the bidding. The transparency and information disclosure, the reserve and final price, and other process outcomes also can be affected (Correia et al., 2020). A conceptual analysis of auction mechanisms can mitigate default in RES auctions (Kreiss et al., 2017) and to guidance for implementation and participation of them (Haufe and Ehrhart, 2018). Prequalification criteria is also one of way to achieve a high realization rate and to ensure the support cost efficiency of the auction scheme (Maurer and Barroso, 2011). In addition to the prequalification criteria, penalties also are identified as an important drive realization rates of auction, while technological banding or the pricing rule are identified as a criteria that do not affect their effectiveness (Matthäus, 2020).

Given that investment in generation often has higher absolute cost implications than transmission, the sequential approach of first defining the generation portfolio, and then the optimal transmission capacity for that portfolio, is, in principle, logic, and often practiced. However, if the location specificity of renewable sources requires further expansion of transmission without this having been envisioned, and priced, even in the long-term planning stage, the initial strategy may result in sub-optimal investment allocation.

An intense discussion can be observed between the operating and centralized planning entities, regarding the use of regional auctions to acquire renewable energy, under the argument that the auctions typically induce a high concentration of renewable energy plants at very productive sites far-off the main load centres, leading to an inefficient allocation as transmission line capacities are restricted but not

considered in the allocation, resulting in an inefficient system configuration in the long run. To counteract these tendencies effectively, some authors propose a combinatorial auction design that allows to implement regional target capacities (Bichler et al., 2020).

Literature also brings a review of auction in various countries of South America in order to compare final prices for technology (Moreno and Larrahondo, 2021) and a guide for designing contracts for RES procured by auctions in Latin America and Caribbean (Correia et al., 2020). The Brazilian experience with wind power in the period from 2009 to 2015 indicates that the level of competition is sufficient to ensure free price formation in the market. The inflation-adjusted auction prices in Brazil decreased by 54% until the end of 2012, before subsequently growing again to 87% of the first-round price. These declines in prices can be attributed to the increasing experience of actors - which number of owners of wind power projects has increased from 16 to 49 actors - and the increasing level of competition among project developers, investors, and turbine manufacturers (Bayer, 2018). The technological evolution of turbines was another important reason that also led to the significant decline in wind energy prices, which brought new perspectives to the source and a clear reduction in unit installation costs.

The synergy effect between a policy mix of Renewable Portfolio Standard (RPS) and long-term contract auctions is suggested to be more significant than alternative policy combinations. It because a technology-specific auction can be a complement to the technology-neutral renewable portfolio standard, not only in terms of minimising risk, but also in terms of cost and dynamic efficiencies (Kwon, 2018).

A broader approach contemplating not only the winner selection process and the results of the auctions but also their institutional and procedural side, with the detailing and specifications throughout the process of contracts and auction design definition, are the main contribution of the paper. The approach of securing the revenue stream and addressing off-taker risk between the sellers and buyers of electricity sector are also an additional contribution.

The next sections provide an outline of the most important auction theoretic elements that found their way into the design of Brazilian auction model.

3. The Brazilian electricity market

The Brazilian power sector was initially dominated by state-owned companies. The state-led model worked well until the second oil shock in 1979 and the deepening of the Brazilian fiscal and hyperinflation crisis during the 1980s. In fact, between 1970 and 1995, installed capacity had grown by 435 per cent. Still, the power companies lost their financial health and investment capability due to the tariff control measures imposed by the federal government to reduce the impact of inflation.

To restore the investment capacity and attract private capital to the power sector, the Brazilian government launched the first institutional reform in 1995, with the following goals (World Bank Group, 2012):

- Electricity generation, transmission, distribution, and trading/marketing should be unbundled into separate segments;
- Electricity generation should become a competitive activity at the risk of the Independent Power Producers (IPP), with prices set by the market;
- Large consumers should be allowed to buy energy freely in the market;
- The transmission utilities should remain as a natural monopoly, with regulation ensuring open access to generators, distribution companies, free consumers and other transmission utilities;
- The distribution companies should also be treated as a natural monopoly and should remain responsible for providing distribution services with open access to free consumers and for buying energy from IPPs to supply their regulated customers;

- An independent regulator would serve as a watchdog for the market, interpreter of specific legislation and guarantor of the stability of rules;
- Boosting supply as an investment opportunity must be left to the market agents.

The institutional reform was only completed in 1997 when Congress approved the legal framework¹ that creates the National Electricity Agency (ANEEL), the Independent System Operator (ONS) and the Wholesale Energy Market Authority (MAE).

According to the initial model, large consumers could sign power purchase agreements (PPA) directly with the IPPs or energy traders. In contrast, other consumers remained supplied by the local distribution company under a regulated tariff defined by ANEEL.

The signed contracts, both by free consumers and distribution companies, had to be registered with the MAE. Any deficit between what was contracted through PPAs and what was consumed needed to be bought by free consumers and distribution companies on the spot market at the short range marginal cost of operation (SRMC) calculated by the ONS.

The ONS was also responsible for the dispatch of power plants (considering the SRMC in a tight pool² approach), planning for the expansion of the transmission grid and granting grid access to consumers and producers.

The institutional model designed in 1995 was not able to guarantee security of supply. In April 2001, the central hydropower plants' reservoir levels had dropped to around 32 per cent of their maximum capacities, with energy deficit risks topping 15 per cent, 10 percentage points higher than the acceptable threshold of 5 per cent.

Although the drought period contributed to rationing, the main reasons for the rationing were that the installed capacity did not follow energy demand growth. The economic signal provided by the spot market in a hydro system with big reservoirs was too risky for investors who depend only on the spot market for revenue. Most of Brazil's electricity is supplied by large hydroelectric plants. Unlike coal, oil or gas plants, their cost of operation is practically zero. Thermal plants play an essential role in complementing the water system, but not in competing with it. While the market would happily accept and pay a reasonable price to produce new natural gas plants in the dry season, a 'wet' year would see little demand for gas generation, and the price paid for electricity would remain at a low level. In short, building gas plants would be like making a climate-based bet for the next decade. Three rainy years in a row and the project would be bankrupt.

The experience built up by Brazil during its rationing crisis in 2001 underscored the need to modify the market design of the Brazilian power sector. The absence of long-term PPAs was too risky for private capital.

On 15 March 2004, the federal government approved Act 10848, which started the second institutional reform of the Brazilian energy system, with four explicit aims (Tolmasquim, 2014): guarantee security of energy supply and resource adequacy in investment; ensure fair and cost-reflective tariffs; reintroduce central planning to cope with demand growth; and build a stable regulatory framework.

Regarding security of supply and the fairness of tariffs, the second reform of the Brazilian energy system promoted (Correia et al., 2006):

- The segregation of the regulated market into one where distribution companies procure energy contracts to supply the captive

consumers, and a free market where IPPs, energy traders and free consumers can transact energy;

- The use of an auction scheme to procure energy for the regulated market;
- The use of long-term contracts in the regulated market to reduce price volatility and provide a predictable revenue stream that can be used as collateral for long-term financing of new projects;
- The obligation of contractual coverage by distributors and free consumers, leaving the spot market only for imbalances;
- The obligation to back all contracts with firm energy (physical coverage³) certified by the Brazilian MME to each power plant. In practice, the contracts must indicate the power plants that will produce the energy and be limited to the firm energy certified by the MME for each power plant. This avoids the situation in which a plant sells more energy through bilateral contracts than it can produce given a certain risk level established by governmental regulation.

The second reform also restored central planning and the role of the MME in the oversight of the sector, with the creation, in 2004, of the Electrical Sector Monitoring Committee, and of the Energy Research Office (Empresa de Pesquisa Energética, or EPE) to support MME with energy planning. The ONS retained the responsibility for short-term planning in transmission.

Finally, to improve the regulatory framework, the reform replaced the MAE with a new market operator (Câmara de Comercialização de Energia Elétrica, or CCEE) with more robust and more transparent governance and under the direct oversight of ANEEL.

The second sectorial reform was successful in creating a buoyant market with an increasing number of agents, especially IPPs, energy traders and free consumers (Table 1).

The energy supply mix had also evolved, from a system strongly dependent on sizeable hydropower plants with a relatively small nuclear and fossil complement, to a system with deep penetration of wind, biomass and solar (Table 2 and Fig. 1).

Finally, Brazil's electricity production and transmission system is a large-scale hydro-thermo-wind system with a predominance of large hydroelectric power plants. Centralized transmission planning ensures the interconnection of electrical systems, enabling energy transfer between subsystems, allowing for synergistic gains, and exploring the diversity between the different hydrological and wind regimes.

4. Regulatory and policy framework

The Brazilian power sector regulatory framework is given by a set of laws, decrees and resolutions issued by the National Congress, Presidency, MME and ANEEL. Over the past few years, the Brazilian electricity sector has undergone structural changes in its regulation. The objective has been the establishment of a model that would promote economic efficiency through the competitive environment, especially in

Table 1
Number of agents in the Brazilian electricity market, various years.

Year	2000	2005	2010	2021
Independent power producers	17	87	290	1416
Energy traders	5	47	93	406
Free consumers	0	470	940	8857
Self-producers	0	14	34	79

Source: CCEE (www.ccee.org.br, accessed May 2021).

¹ Acts 9074, 9427 and 9648.

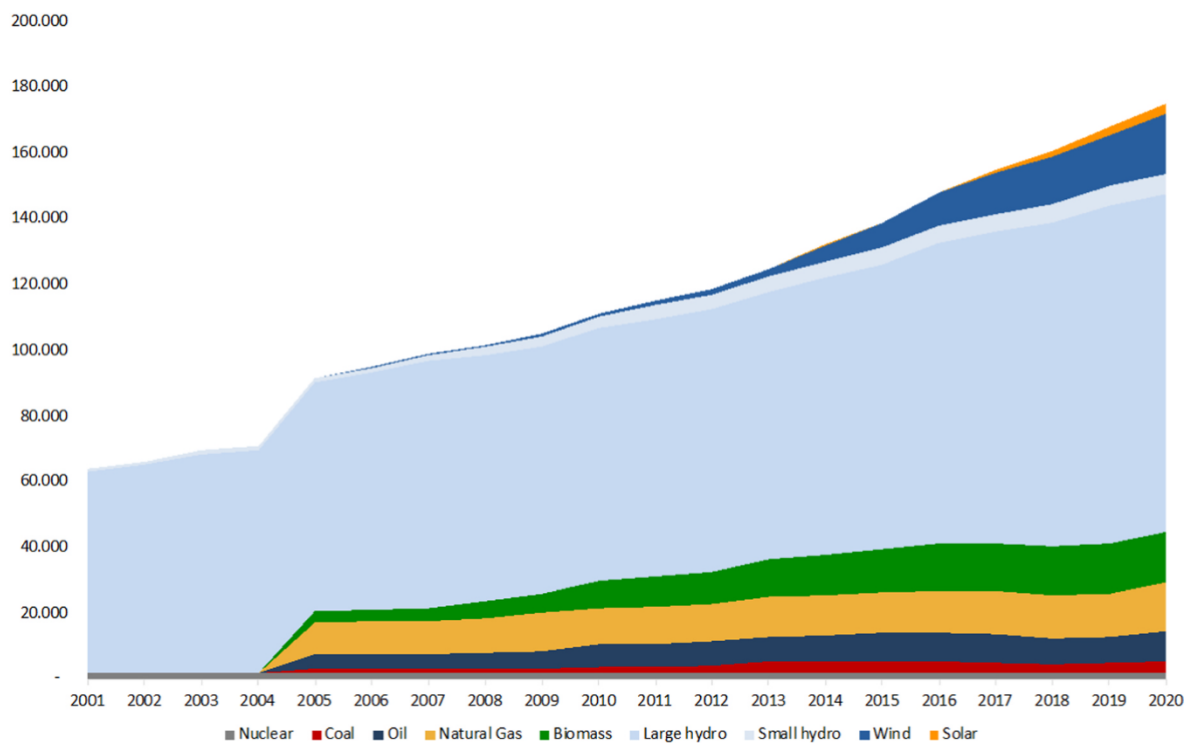
² In the tight pool model, the dispatch is centralised and based on pre-determined variable costs. For comparison, in the loose pool model, the dispatch is centralised, but the generators are free to offer any price they like; such an approach is also referred to as price-based pools. Finally, the dispatch can be decentralised and entirely based on price.

³ It refers to the expected generation of energy that the power plant will be able to provide under critical conditions, especially regarding the seasonality and variability of RE sources.

Table 2

The Brazilian electricity sector, December 2020.

	GW	%
Total capacity	177.16	100
Large hydro	102.99	58.1
Wind	18.02	10.2
Gas	14.83	8.4
Biomass and waste	15.28	8.6
Oil	9.09	5.1
Small hydro	6.33	3.6
Coal	3.58	2.0
Solar photovoltaic	3.33	2.0
Nuclear	1.99	1.1
Others	1.72	0.1
Urban electricity access rates	99.96	%
Rural electricity access rates^a	98.20	%
Peak demand	85.97	GWh/h

Source: ANEEL (www.aneel.gov.br, accessed May 2021).^a This value includes both on-grid and off-grid supply solutions.**Fig. 1.** Installed electricity generation capacity (MW), 2004–2020Source: ANEEL (www.aneel.gov.br, accessed May 2021).

the generation sector, which would make investments in the expansion of installed capacity feasible and guarantee service to the consumer market.

Brazil's government determined that it was necessary to have an institution based on knowledge and technical excellence, with permanent high-level professional staff, tools, and a database suitable for the formulation of energy policies and decision support. EPE deserves special mention as one of the most important institutional innovations of the new regulatory framework, mainly responsible for energy planning (energy expansion and transmission facilities). The EPE furthermore provides a range of analyses and reports on energy statistics, energy efficiency and socio-environmental studies (including environmental feasibility and sustainability of electricity and gas production and transmission sources, energy resource inventory and prospection for the preliminary environment licensing process of strategic hydropower and transmissions projects).

4.1. Renewable energy auctions

The Brazilian auction programme was created by Act 10848 with the objective to provide an efficient, transparent, and competitive instrument for the awarding of long-term PPAs for captive consumers supplied by the distribution companies. Since 2005 all the energy contracts in the regulated market have been secured through an auction scheme prioritising the procurement of greenfield projects to meet demand growth.⁴ The auctions have been designed to provide long-term contracts to new power plants and facilitate their financing through project finance, where lenders provide loans based on the projected cash flows of the project rather than on the balance sheets of its sponsors.

The electricity commercialisation contract in the regulated market is a bilateral contract for the purchase and sale of electric energy and its associated capacity, signed between the selling company and the distribution company within the scope of the regulated market, as a result of auctions of electricity from existing generation plants and greenfield plants. In other words, there are not any separate tenders to contract

energy and capacity in Brazil. This is mainly due to the large number of hydropower plants within the power system deployed with excess capacity to allow power demand to be met throughout the dry season and critical wet periods. Therefore, energy contracting ensured the availability of capacity. With the increase of run-on-river hydropower plants and variable renewable sources, capacity is becoming an issue and the government has started discussing the possibility of running capacity-only auctions.

There are five types of energy auctions:

Greenfield auction: The greenfield project auction aims to meet the increase in distributors' power demand by contracting energy from plants that have yet to be built. This auction can be of four types: A-6

⁴ Originally, the auctions for new projects had a lead time of 5 or 3 years and PPAs lasted between 15 and 30 years.

(plants that go into commercial operation in up to 6 years' time), A-5 (in 5 years), A-4 (in 4 years) and A-3 (in 3 years). The winners of the auction sign contracts with the distribution companies that are procuring energy. Only the consumers from the regulated market pay for this energy. The Brazilian government has implemented 31 rounds of greenfield project auctions, with 30 rounds specifically for RES.⁵

Reserve auction: The contracting of reserve energy was originally created to increase security in the electricity supply in the National Interconnected System, with power from greenfield or existing plants specially contracted for this purpose. So far, reserve auctions have only contracted greenfield plants. However, the term reserve energy does not have the traditional connotation of power that acts as a system backup. This is because renewable sources, mostly contracted in these auctions, are non-controllable sources and have inflexible dispatch. Thus, the concept of reserve energy is here interpreted as additional energy to the system that aims to reduce the depreciation of hydroelectric power plant reservoirs. CCEE act as a single buyer and sign contracts with all the winners of the auction. Reserve energy is accounted for and settled on the short-term market operated by the CCEE. This type of 'insurance' in the energy supply generated the Reserve Energy Charge intended to cover these costs – including administrative, financial and tax costs. As the reserve auction works as an insurance for all the power system, the energy charges apply to all the consumers from the regulated and free market. The Brazilian government has implemented 9 rounds of reserve auctions, the last in 2017.

Renewable energy auction: The auction of renewable sources was instituted to meet the growth of the market in the regulated environment and increase the share of renewable sources – wind, solar, biomass and energy from small hydroelectric plants – in the Brazilian energy system. In the last years, the government has been using only greenfield auctions to procure new RE plants. The Brazilian government has implemented 3 rounds of special auctions for RES, the first in 2007 and the last in 2015. Despite the creation of a specific auction modality to procure renewable energy, most RE capacity was procured through "greenfield auctions" and "reserve auctions".

Existing power plant auction: This auction was created to contract energy generated by plants already built and in operation, whose investments have already been partially or fully amortized and, therefore, have a lower cost. The separation of the existing and greenfield energy auctions allowed the average price in the regulated market to be calculated apart from the marginal expansion cost, thus contributing to fair electricity rates. Moreover, this offered distribution companies the possibility of signing agreements with the existing generation companies for shorter lead times and duration as a risk management tool designed to offset uncertainties in demand and the loss of free consumers. Finally, this separation prevented the existing plants from squeezing new plants out of the picture in the regulated market, thus helping ensure energy security. In total, 21 rounds of existing power plant auctions have been implemented, 19 with specific products for RES.

Adjustment auction: This auction aims to adjust the distributors' energy contracting, addressing any deviations arising from the difference between forecasts made by distributors in previous auctions and the actual demand growth. As a result of this auction, the distribution companies sign short-term contracts (from 3 months to 2 years) with the auction winner. Seventeen rounds have been implemented since 2005, none with specific products for RES.

Auctions to procure greenfield power plants, reserve energy and RES usually have a lead time to commercial operation of between 3 and 6 years. Project developers are awarded 15 to 30-year fixed tariff contracts (Fig. 2).

In short, the new regulatory framework was designed to foster vast amounts of investment in the generation capacity expansion necessary

to meet a fast-growing demand at the lowest cost possible (Tolmasquim, 2014). Table 3 presents the key auction information.

4.2. Auction design

The energy auctions in Brazil are conducted annually according to a schedule released by MME. The auction process is led by ANEEL under the guidelines of the MME. An auction committee undertakes the main auction tasks, which are distributed among different institutions (EPE, CCEE, ANEEL, MME). Once an auction is concluded, the winning generator companies sign contracts directly with distribution companies, or with the CCEE in the case of a reserve auction (Fig. 3).

The auctions have three stages: the first for registration and technical prequalification; the second for bid submission and the winners' selection; and a final stage for the legal and financial qualification of the preferred bidders, performed by the regulatory agency (Fig. 4).

The first and third stages aim to reduce the risk of failure in the implementation of the project, and they do not intend to rank the projects. The assessment of the required documents in stage 1 is proactive to ensure the participation of the largest possible number of projects in the bidding phase (second stage). However, considering the large number of projects registered in Brazil every auction, the qualification process was split, and the legal and economic documents of preferred bidders are evaluated in stage 3 to reduce transaction costs for the auctioneer. The second stage encompasses the bidding, and the winner selection processes.

Since energy auctions have been regularly performed in Brazil, investors already have a pipeline of projects planned and there is no need for a long notice period to respond to the first-stage request for registration. Developers thus normally have a 30-day period during which they must respond to the call for registration. The EPE has 80 days to evaluate the projects and documents presented and to calculate the firm energy of each project.

In parallel, ANEEL opens a public consultation process for stakeholders to analyse the rules of the auction⁶ and the draft contracts. Bidders have access to complete information about the auction, including the final version of the rules and contracts, the price cap, and the certified firm energy requirement (physical guarantee). The information about the total demand for the auction is not disclosed in advance to mitigate the risk of collusion.

Winners must secure all permits and licences, reach financial closure, complete the construction of the power plant, and connect it to the grid within the A-X period stipulated in the rules (Fig. 2). During the construction period, the investor can change some of the technical characteristics of their project. According to the rules of Brazil's A-6⁷ auction of 2019, for example, it was possible to change the installed capacity, the turbine type and quantity of generating units, and the connection point of the power plant. Technical changes must conform to environmental permits, cannot modify the energy source initially indicated, and may not compromise the fulfilment of contractual obligations assumed in the auction, such as the amount of energy and capacity negotiated, the date of commencement and the duration of the contract. Bidders in the 2019 auction were also allowed to complete power plant construction early and sell energy in the Brazilian wholesale market. The auction rules made it clear that all risks and costs associated with changes in the technical characteristics and the plant's commercial operations date were the exclusive responsibility of the seller and could not be passed on to the buyers.

Energy auctions and contracts have evolved over the years. In the contracts for different sources and products, different delivery obligations are designed, such as the determination of wind or photovoltaic

⁵ RE includes all renewable energy sources, including large hydropower plants.

⁶ Including the detailed auction process, the price rule and the winners' selection criteria.

⁷ A-6 refers to the lead time of 6 years.

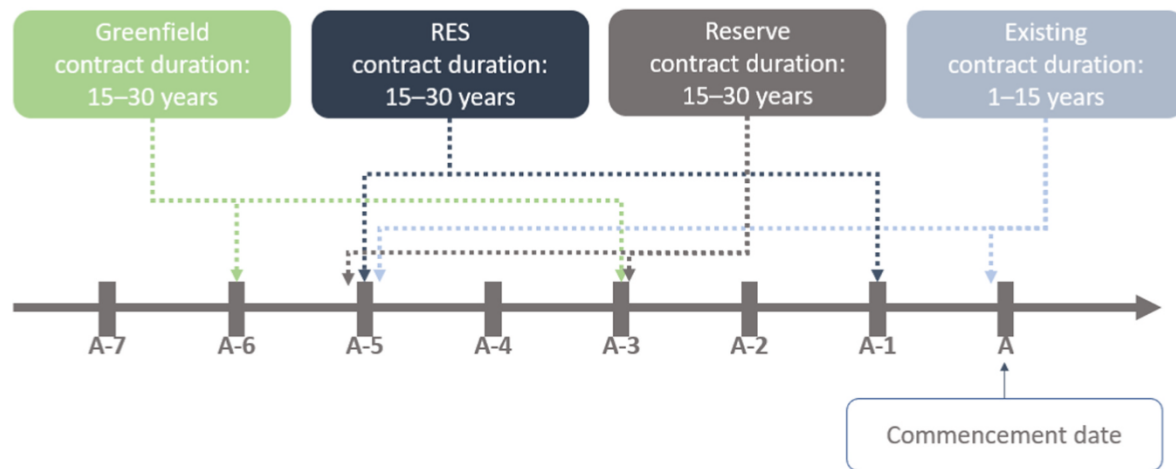


Fig. 2. Lead time to commercial operation of the auction projects.

Table 3
Key auction information.

	Year of introduction	2005
Design	Frequency of auctions/rounds	82 rounds (61 rounds with specific products for RES) ● 40 rounds for existing power plants, 21 with specific products for RES^a ● 30 rounds for greenfield projects, including the special rounds for the hydropower plants of Santo Antonio, Jirau and Belo Monte (28 rounds with specific products for RES) ● 3 rounds for only RES power plants ● 9 rounds for reserve energy (all with specific products for RES)
	Currency	Brazilian Reais (indexed to local inflation)
Implementation	Policy and regulation guidelines	Ministry of Energy and Mining
	Regulator	ANEEL
	Procurer	CCEE delegated by ANEEL
	Off-taker	Distribution companies and the CCEE, in the case of reserve energy
Outcomes	New MW procured ^b	105228 MW (76862 MW from RES)
	Technology procured	Oil, coal, natural gas, wind, solar, biomass, small hydro, and large hydro

^a Only large and small hydropower plants.

^b Greenfield, RES, and reserve auctions.

energy generation on an annual and 4-year basis. Risk allocations are also distinct between auctions. In reserve energy auctions the risk of generation insufficiency is allocated to all consumers while in new energy auctions (greenfield, RES) the risk is allocated to generators. Table 4 shows the Brazilian auction basic design in 2019.

The contracting is carried out through Contracts for the Sale of Energy in the Regulated Environment (CCEAR) in the modality quantity contract or availability contract, signed between each winning generator and the distributors with demand declarations in the auction. Availability contracts were used to contract energy from RES until 2017, whose risks could be better predicted and priced. However, considering that renewable sources, especially wind, already have a greater forecasting capacity and technological maturity, since the 2018 auctions, contracts have been signed in the quantity modality to mitigate the risk of consumers to exposure of price in the short-term market. Even so, quantity contracts allow the generator to allocate the amount of monthly energy following the load profile declared by the buyer, but with

minimum and maximum limits between 85% and 115% of the annual average of contracted energy, respecting the limits of the installed power of the enterprise.

4.2.1. Bidder ranking and winner selection

The regular auction procedure is a hybrid model that occurs in two phases. The first phase is a descending-clock auction, or Dutch auction, of uniform price, that starts with a ceiling price defined by MME (Fig. 5). Bidders indicate how much they are willing to supply at this price. The auctioneer then lowers the price until the desired supply level is met, plus a certain margin. The auction uses an inflated demand level to stimulate competition in the second phase.

Phase 2 operates as a final pay-as-bid round for the winners of phase 1. Remaining bidders offer a final sealed price, which cannot be higher than the price disclosed in phase 1. The clearing price is determined when supply equals demand, and the winning bids are those lower than the clearing one (Fig. 6). For simplicity, winner selection under the Brazilian auction scheme is based exclusively on price.

Due the increase in construction timelines of transmission lines in Brazil and the uncertainty about the auction's winners' location, since 2013, auctions with a lead time shorter than 5 years have been performed with a different approach. This new design includes a preliminary phase to select the projects with connection feasibility through competition among projects with the same connection point. In the preliminary phase, bidders thus submit a single bid with price and quantity for each project. The auctioneer classifies bids according to their price at each connection point following the discriminatory price methodology (Fig. 7). Projects that exceed the 'transmission margin', calculated by EPE and ONS for each connection point indicated by the registered projects, are excluded from the first and second phases of the auction. The winners then engage in the general auction process following the methodology described in Figs. 5 and 6.

From 2017, the Brazilian auction programme again changed the bidding process. The new methodology is called a continuous trade reverse auction. The new design also has 2 phases, but in this first phase, bidders submit sealed bids with a price and quantity. Bids are evaluated by price, and the auctioneer classifies the 'temporary winners' as the lowest bids up to the market-clearing quantity (demand), while all other bidders are considered temporarily disqualified. The second phase is a descending clock iteration of 3 or 5 min (depending on specific auction rules), in which any temporarily disqualified bidder can replace a temporary winner by submitting a bid lower than the marginal price minus a decrement. The decrement (the minimum difference between the marginal price of temporary winning bids and the new bids) is set by the auctioneer prior to the auction (Hochberg and Poudineh, 2018).

The price presented in the first phase (during the transmission point

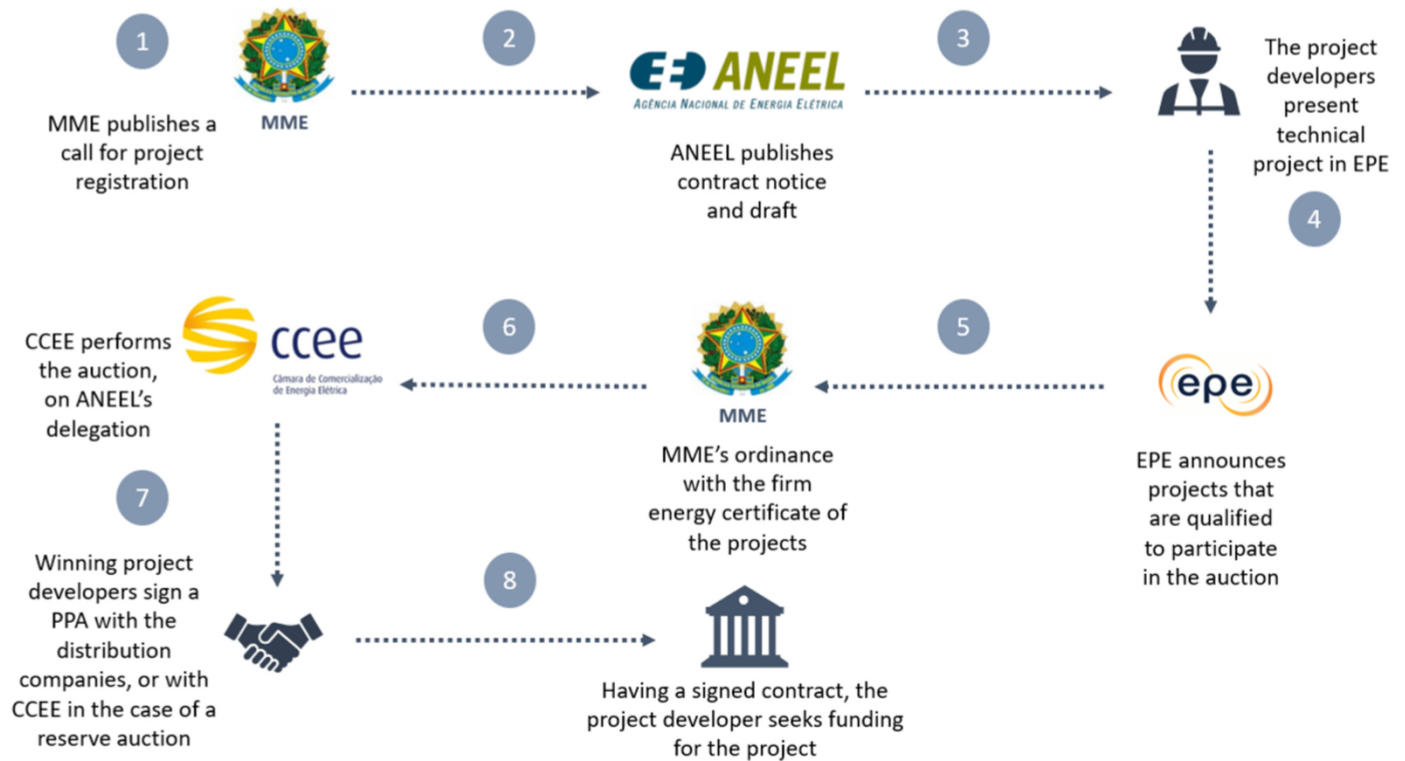


Fig. 3. Auction process overview.

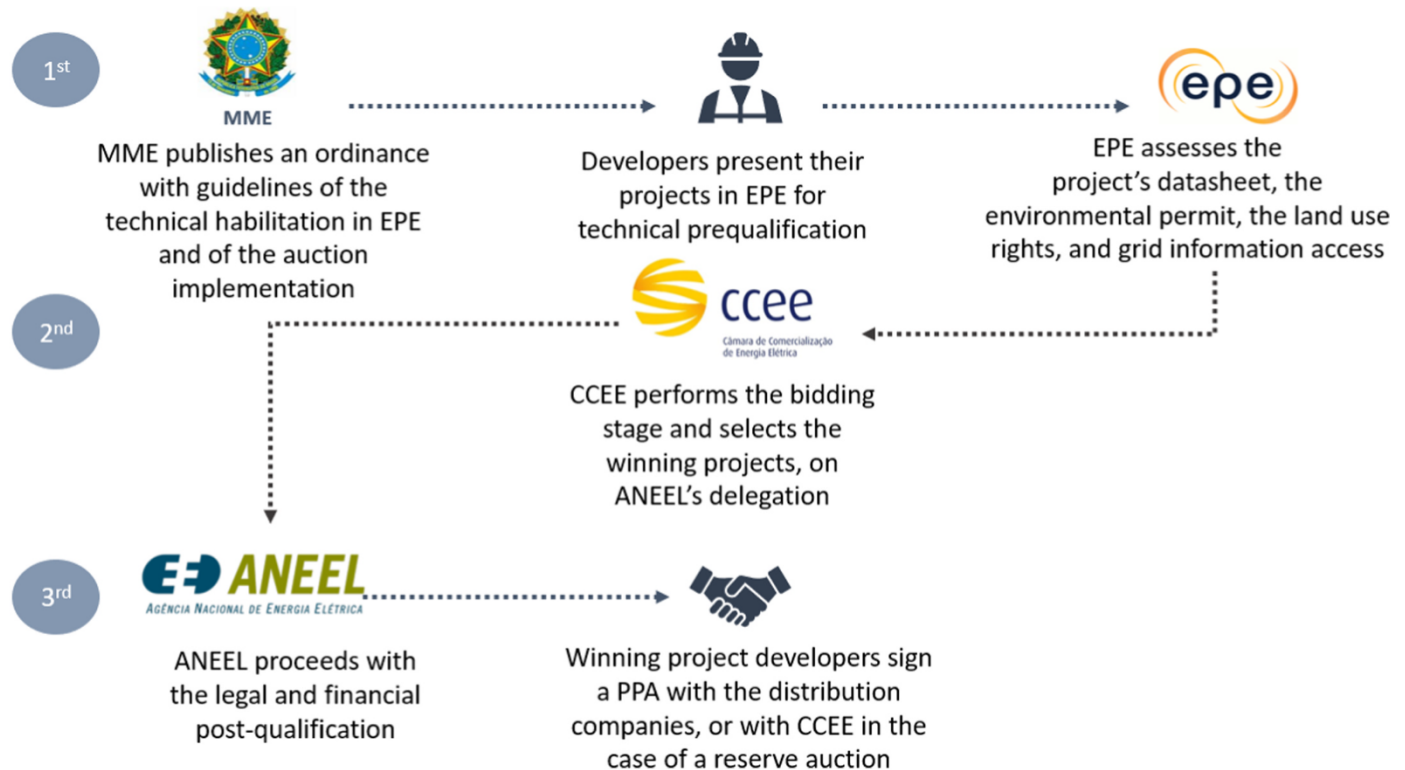


Fig. 4. Auction stages.

competition) cannot be increased in the second phase, and the bidders remain limited to their prior bid (price and quantity) until the current bidding price of the second phase reaches their price. At that moment, the bidders may submit a new bid with a lower price or remove their

offer from the auction. The auction closes when the total supply has met the auction's demand, and the outcome can be a mix of uniform and discriminatory price, depending on how fierce the competition for the first phase was (Fig. 8).

Table 4
The Brazilian auction basic design.

Periodicity	Annually – usually 2 rounds per year
Project preparation	Bidders must secure social and environmental permits, land use rights and interconnection agreements to be allowed to register for bidding
Stages	
Registration and qualification	Evaluation of the required permits and documents
Bidding	Hybrid price rule – pay-as-bid with uniform pricing (highest accepted bid)
Auction demand	
Greenfield projects	Decentralised – Distribution companies
RES	Decentralised – Distribution companies
Reserve	Centralised – MME
Energy source	Technology-specific or group of technologies
Winner selection	Only price
Lead time	Between 3 and 6 years
Risks	
Seller	Construction, operation, equipment performance risks and exposure to the spot market (in the case of the contract for energy – ‘quantity contract’)
Buyer	Inflation and exposure to the spot market, in the case of the contract for availability
Liabilities	
Seller	Bid bonds and surety and performance bonds
Buyer	Payment financial warranties

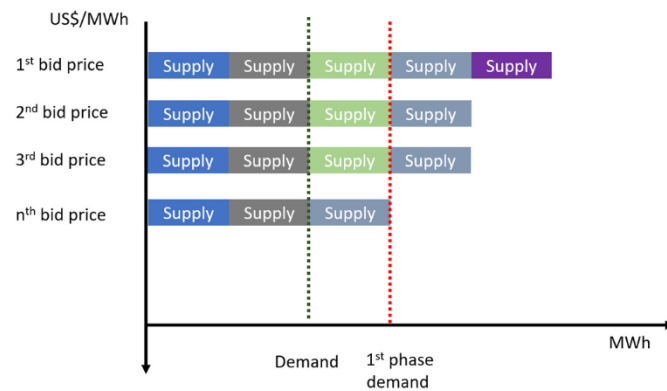


Fig. 5. First phase: Uniform-price auction.

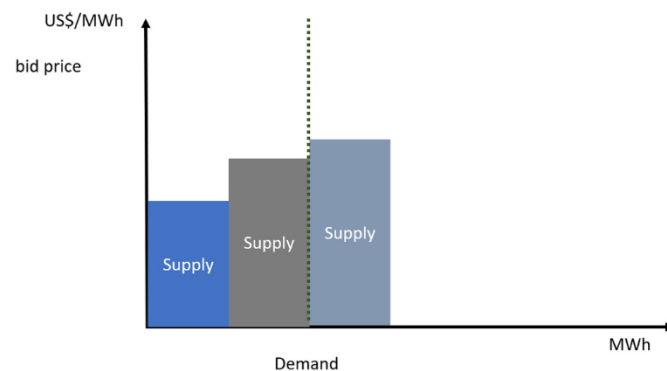


Fig. 6. Second phase: Pay-as-bid auction.

Moreover, considering that the supply is presented by projects that must be entirely constructed, if selected as winners, the amount of contracted energy associated with the marginal project may be greater than necessary to meet demand.

In these cases, until 2018, the supply quantity of the marginal project was fully contracted, and the cost of over contracting was passed on to consumers. However, as of Auction A-6 of 2018, the Brazilian auctions

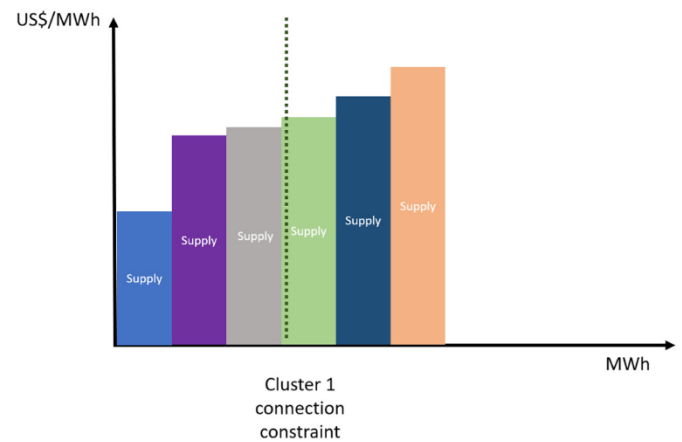


Fig. 7. Preliminary phase for transmission dispute: Pay-as-bid auction.

began to count on the possibility of partial contracting of the marginal project, which could accept the sale of a reduced quantity or withdraw its offer, in which case the demand is not fully met. Since then, the inclusion of this bid ratification phase in auctions has been decided in a discretionary manner, based on the perception of the risk and cost of over- or under-contracting.

4.2.2. Securing the revenue stream and addressing off-taker risk

Investor confidence can be significantly improved through the provision of payment guarantees by off-takers, resulting in more competitive offers and higher realisation rates. In the Brazilian scheme, the energy may be contracted by distribution companies or procured directly by the MME, in the case of reserve energy. In the first case, total demand is aggregated, and each preferred project signs off-take contracts with each distribution company, thereby reducing payment risks through a portfolio effect. Moreover, the distribution companies include the energy cost of the distribution tariffs and transfer it to the regulated consumers. In the second case, reserve energy is funded by an energy levy that is collected among free and regulated consumers.

Brazil uses a portfolio of off-takers backed by accounts receivable to reduce the investor's risks. In this mechanism, the off-taker of the energy auctions is the pool of all electricity distribution companies. Each auction winner signs a different contract with each distribution company in the pool. Thus, only a fraction of the energy is committed in each agreement, mitigating default impact on investors. At the same time, the amount of energy acquired by each distribution company is divided among several projects, mitigating the risk of delay for the buyers.

To secure the revenue stream and to limit off-taker risk, the distribution companies must provide an accounts receivable assignment to the IPP, meaning that revenue collected from end-users will be deposited into a specific bank account, under the supervision of a managing bank, which will ensure that energy purchase contracts will be paid before the funds are made available to the distribution companies (Fig. 9).

Finally, considering the complexity of the contracts and the market uncertainties, the Brazilian contracts establish arbitration as the primary mechanism for dispute settlement. The arbitration instrument is usually faster than the judicial and administrative courts and ensures that the arbitrators will be experts.

4.3. Auction implementation

Auction success, in general, is as much dependent on good auction design as on bidder trust in the auction process. The presence of a mandated, credible, well-capacitated and well-resourced agency responsible for managing and implementing the auction process is a critical success factor (Eberhard et al., 2014; Eberhard and Naude,

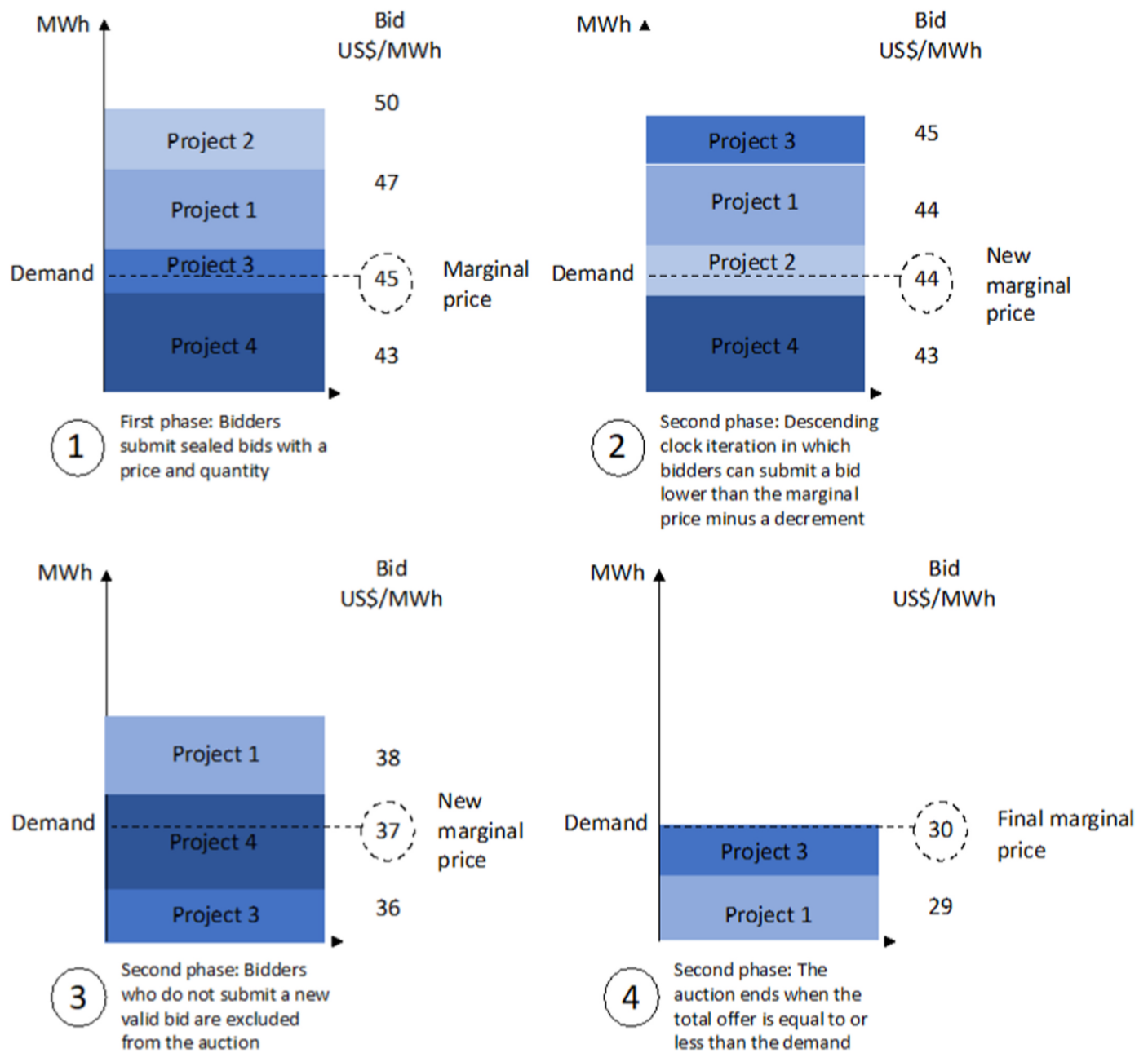


Fig. 8. Continuous trade reverse auction (since 2017).

2016). Coordination among government entities is also essential to the success of the auction. The inputs of various government departments and agencies impact the process of project prequalification, mainly because developers need to secure preliminary environmental permits and land use rights, which means a long and difficult process of engagement with federal, local, and state-level authorities.

In the Brazilian scheme, the auction programme is implemented by four institutions. The MME is responsible for setting the guidelines, the schedule, and the design of the auction, including bidding rules and winning selection criteria, and for issuing the call for projects registration.

The EPE is responsible for the prequalification stage, assessing the registered projects' data-sheet and the documentation for land use rights and environmental permits. The information collected by EPE during the prequalification process is used to prepare a technical note on the price cap of the auction and the maximum amount of energy each project will be allowed to sell. The EPE technical notes support the MME

decision about those issues.

ANEEL prepares the draft contracts, performs the auction with the technical support of the CCEE, conducts the technical and financial qualification of the preferred bidders, and oversees the signature of the contracts and the completion of the projects in due time. The energy contracts are signed by the distribution companies and the reserve contracts by the CCEE.

The role of each institution is determined by Decree 5163 but reflects the division of responsibilities established by the Brazilian regulatory framework, especially by Acts 9427 and 10848, that creates ANEEL and regulates the production and procurement of energy to the regulated consumers.

The designed process (Fig. 10) is complex and time-demanding but has been performed with a high level of transparency (every auction round is preceded by a public consultation), and predictability, contributing to an increase in bidder confidence in the process and in political support.

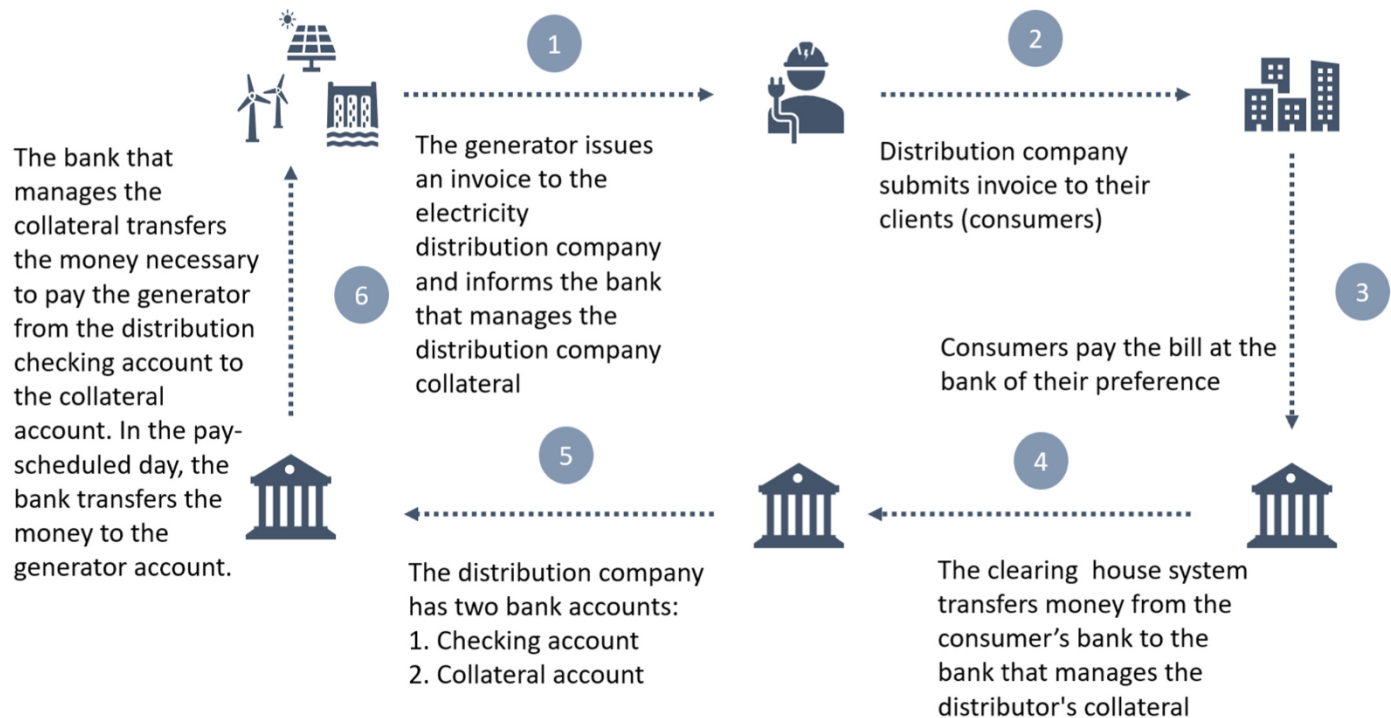


Fig. 9. Accounts receivable scheme.

The auction programme is funded by the buyers and sellers proportionally to the contracted energy. If the auction is cancelled or performed without the sale of energy, the auction costs will be paid entirely by the buyers, in proportion to their announced demand. The cost of each auction is, however, low and covers only the expenditures of the CCEE, since MME, ANEEL and EPE are funded by public budget. The total cost of the 30th greenfield auction, performed on 18 October 2019, was, for example, US\$ 13000.

The auctions are performed electronically using a platform developed by the CCEE. The integrity of the process, the outcomes and the security of the information exchanged during the auction are audited by an independent third party.

5. Results

Brazil's regulated auctions contracted 9.571 TWh of energy between December 2004 and October 2019, of which 8.180 TWh was from new (greenfield) power projects, adding 105.2 GW (76.8 GW of RES) to the grid (Figs. 11 and 12).

The auction scheme also fostered the growth of non-conventional renewable sources in Brazil. This process was accompanied by an initial increase in the price of biomass and small hydro until December 2009, when competition from wind power plants reversed the trend. The competitiveness of wind technology is partly explained by the worldwide reduction of the equipment costs due to technological advances and economies of scale, but also reflects the development of local investment capacity. Additionally, the period was characterised by a gradual reduction in the cost of capital in Brazil.

In 2013, however, the wind price trends reversed again, and wind energy prices returned to the initial US\$55/MWh level in November 2015, while the biomass price reached a peak of US\$89/MWh in April 2015 (Fig. 13). Two effects contributed to the price hike. First, the growth in the number of projects facing construction delays and the bankruptcy of a large local equipment provider increased investors' risk perception. Second, the return of inflation acceleration in Brazil also reversed the downward trend in the basic interest rate (SELIC), making financing scarcer and costly.

Finally, the entry of solar PV energy in the market and the return of the interest rate reductions from 2015 gave new impetus to the reduction of energy prices in Brazil (Fig. 13).

On the other hand, the emphasis of the Brazilian scheme on procuring new power plants implies a residual risk of delay or bankruptcy. Figs. 14–16 present the situation of 374 power plants in construction in October 2019 and show the capacity (in MW) deployment to comply with the contracted timetable. About 7 per cent of solar PV, 30 per cent of wind and 50 per cent of biomass power plants are delayed. More relevant, 14 per cent of wind and 12 per cent of biomass power plants do not even have a probable date of completion, indicating a strong likelihood that the contracts will be terminated.

The leading causes of delays are construction cost overruns (18 per cent), including the impact of the exchange rate variation; the unanticipated difficulty of obtaining financing (26 per cent); difficulties in obtaining construction, operation, and environmental permits (18 per cent); delays in connecting to the transmission system (20 per cent); and problems with equipment suppliers (12 per cent).

To reduce the environmental risk, the Brazilian auction scheme has a prequalification phase where the investors must present, at least, a preliminary environmental permit. However, some features of the Brazilian environmental licensing process make it very long, complex and uncertain, and securing the preliminary permit does not guarantee that the construction and operation permits will be issued. Indeed, a report by the legislative consultancy of the Brazilian Federal Senate pointed out the main flaws of the environmental licensing process (Hofmann, 2015):

- The environment impact assessments and reports' extensive focus on the negative impact of the project, ignoring the positive externalities;
- The excessive imposition of conditioning factors and mitigating actions by the public authorities;
- The multiplicity of actors with discretionary power;
- The frequent judicial control;
- The absence of a strategic environmental policy;
- The scarcity of systematised environmental data and public information; and

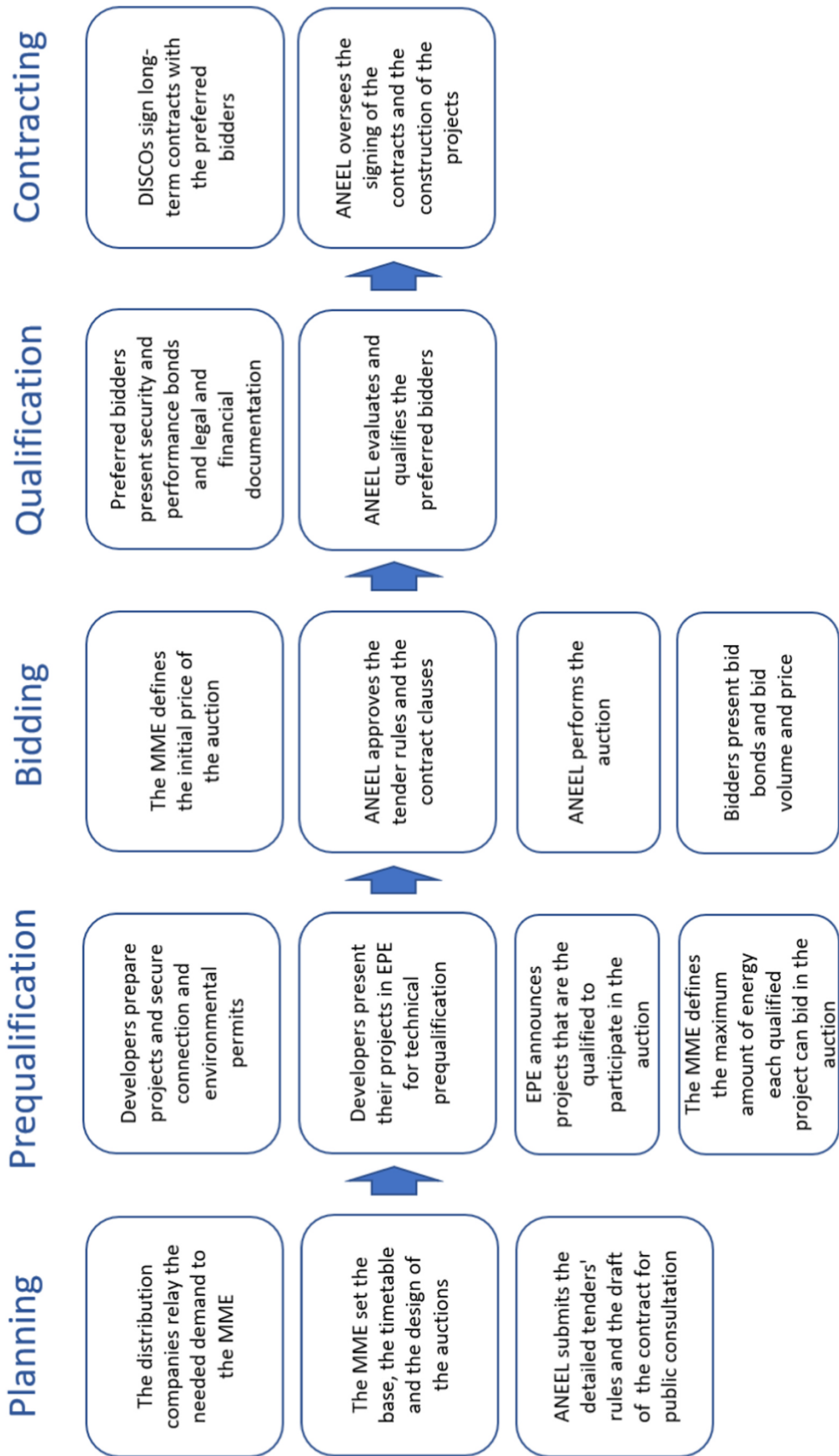


Fig. 10. Brazilian auction implementation scheme.

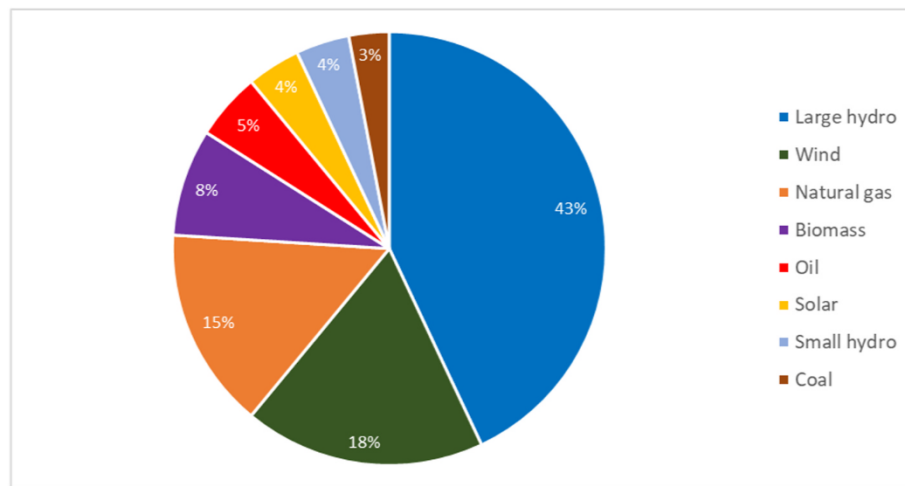


Fig. 11. Brazilian auction outcomes: Contracted capacity. Source: CCEE (www.ccee.org.br, accessed June 2020).

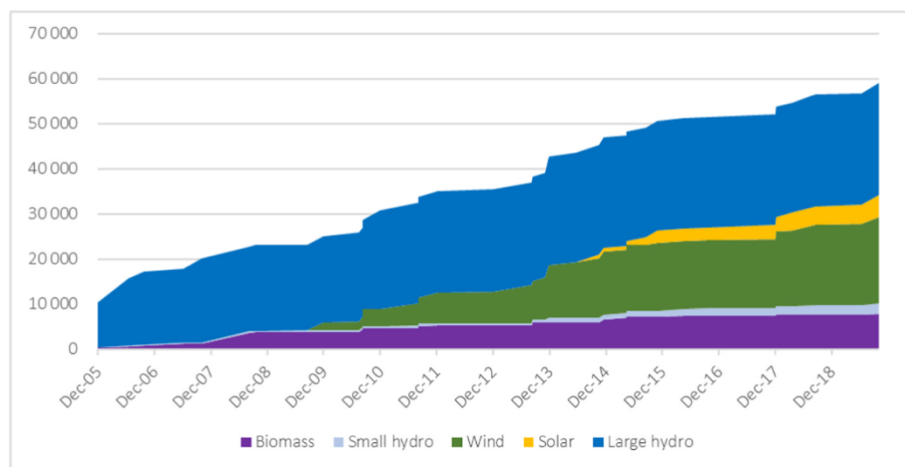


Fig. 12. Brazilian auction outcomes (MW): New capacity of RES. Source: CCEE (www.ccee.org.br, accessed June 2020).

- The excess of bureaucracy, and the use of vague terms such as ‘low environmental impact’ and ‘directly or indirectly affected area’ in the resolutions and guidelines provided by CONAMA and MMA.

Finally, despite following all the rules, the environmental, construction and operation permits are not bonded. Thus, any public authority involved in the licensing process may request further studies and revise its previous understanding regarding the permits at any time, especially if an environmental or archaeological finding is identified. Thus, environmental risk in Brazil is very relevant and difficult to mitigate.

Transmission connection challenges were the result of the previously discussed approach to transmission system expansion planning and tendering, which initially would only commence after energy auctions were finalised.

Finally, the company IMPSA founded Wind Power Energia (WPE) in December 2006 to produce wind and hydroelectric equipment in Pernambuco. At the end of the first semester of 2014, the economic and financial situation of WPE began to deteriorate. After delays in the payment of international debts, on 5 December 2014, WPE requested judicial reorganisation to avoid bankruptcy. The immediate outcome was the interruption of the delivery of all contracted equipment, amounting to an installed capacity of 1580 MW, which resulted in the

construction delay of 26 projects totalling 579.2 MW (ANEEL, 2016).

6. Lessons learned and recommendations

The Brazilian auctions have been successful in scaling up private investment and the deployment of RES at fair prices, confirming that auctions are effective and efficient at revealing prices under uncertainty (Haufe and Ehrhart, 2018; Hubbard and Paarsch, 2016). Nevertheless, contracting power plants in the project phase involves risks related to cost overruns, construction delays and price volatility that cannot be completely mitigated. Brazil’s experience highlights important lessons on how these risks can be dealt with, both in terms of how auctions are implemented as well as designed.

Effective and efficient auctions start with a comprehensive regulatory framework. The responsibility of each institution that takes part in the auction programme must be clearly defined and understood by all stakeholders. The institutions must act in cooperation and under the coordination of a single organisation that clearly defines the role and the boundaries for each institution. Second, institutions must be adequately funded and provided with human resources.

Participating in auctions has significant transaction costs, including the cost of preparing and developing the project, obtaining permits, and providing bid bonds and other financial warranties. If auctions are not

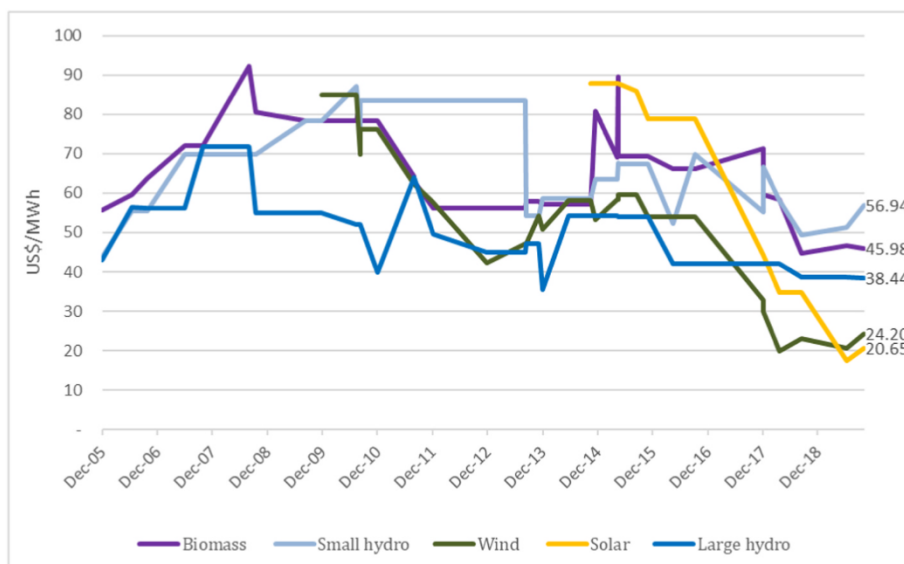


Fig. 13. Brazilian auction outcomes: PricesSource CCEE (www.ccee.org.br, accessed June 2020).

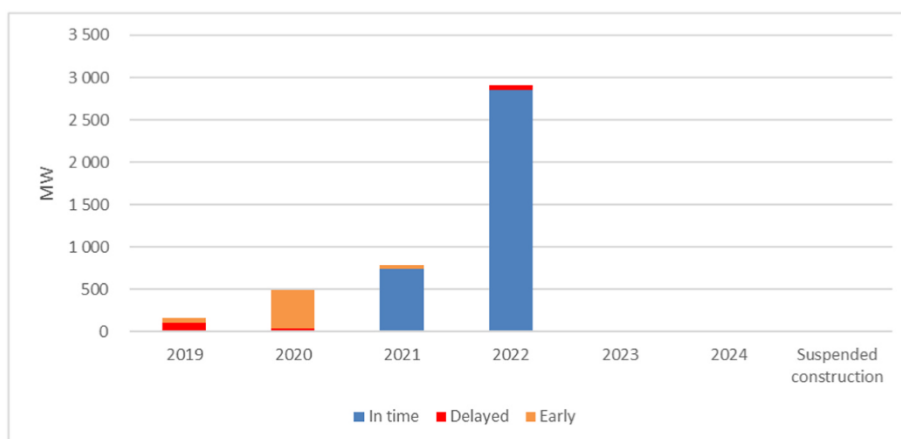


Fig. 14. PV power plants: Compliance with construction timetable Source: ANEEL (www.aneel.gov.br, accessed June 2020).

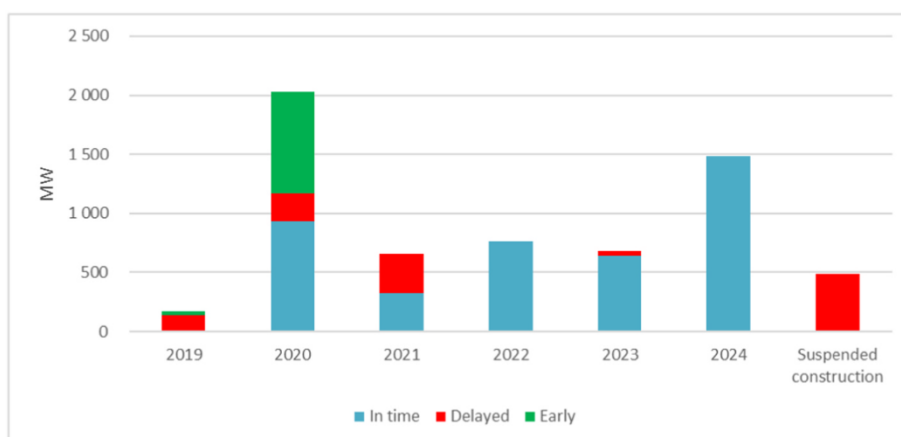


Fig. 15. Wind power plants: Compliance with construction timetable Source: ANEEL (www.aneel.gov.br, accessed June 2020).

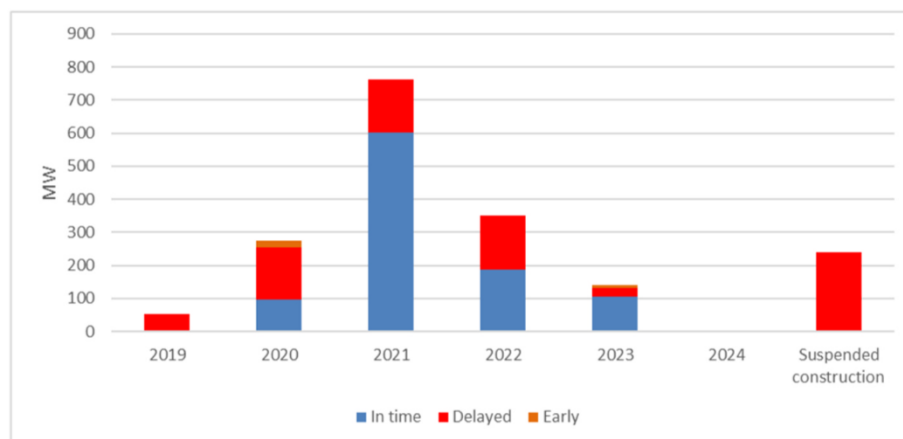


Fig. 16. Biomass power plants: Compliance with construction timetable
Source: ANEEL (www.aneel.gov.br, accessed June 2020).

part of a periodic and predictable process, the risk of potential bidders is amplified, reducing their participation and, therefore, competition in the auction, so increasing the final price. Therefore, auctions should be as recurrent, transparent, and predictable as possible. The preparation and disclosure of a medium- and long-term expansion plan assists greatly in this regard. Additionally, the high frequency and predictability of the auctions, combined with financing support from Brazilian Development Bank (Banco Nacional de Desenvolvimento Econômico e Social, or BNDES), reduced the investment risk for equipment manufacturers and construction service providers, increasing local content in the power plants without the need for specific requirements stipulated in the auction rules.

Developers and investors also need enough time to engage with the auction programme. Lead times must correspond to the longest time necessary to develop the project, secure the necessary permits and licences and deploy a new power plant.

The contract duration should also be long enough to provide investors with some cash-flow stability and predictability during the time required for loan maturity. Consequently, the contract duration should reflect the capital intensity of the technology, the principles of project financing and the discount rate involved.

The use of a portfolio of off-takers (aggregated demand of distribution companies) backed by an accounts receivable mechanism significantly mitigates payment risk and improves the developers' ability to access finance.

Auctioned contracts must be as straightforward and concise as possible in order to avoid the ambiguity that may result in future litigation, but must also allocate the most common risks, limit exposure of each party involved, and clearly outline the procedures to be followed in the event of a claim.

The Brazilian experience also demonstrates that the auction programme can be used to address multiple goals with the use of complementary auctions with differences in the lead time, contract duration and contractual obligations. The auction may demand specific renewable sources or multiples sources (mix of technologies or even technology neutral), but each energy source has an individual contract, specially designed to conciliate the need of the power system and the features of different energy sources.

Another challenge to auction design is to mitigate the winner's curse risk, understood as the failure of the investor to measure all project risks and costs (Krishna, 2002). The winner's curse cannot be entirely avoided but there are some approaches to the design of the auction and the contracts that can mitigate its effects (Correia et al., 2020). First, the auction rules, the draft contracts, and all the information necessary for the investor to elaborate their bidding strategy must be made available well in advance. Second, a qualification phase mitigates part of the risk,

given that the bidders must prove the feasibility of the project by presenting environmental permits and financial guarantees. Third, the auction's design may help to reduce the winner's curse risk. The time and money spent to prepare the projects and participate in the auction are a sunk cost and may influence bidders' strategy. The higher the sunk cost, the greater the risk aversion of losing the auction, and the greater the aggressiveness of the bidder. Aggressive bidders think that it is more important to not lose the auction than seize a small gain of slightly increasing the bid. Consequently, the auctioneer may expect lower prices under a pay-as-bid sealed auction than under a pay-as-clear auction, but the trade-off might be a higher risk of the winner's curse in this scenario. On the other hand, a pay-as-clear process might extract less surplus from the seller and result in slightly higher prices but reduces the risk of the winner's curse.

The remaining risk must be addressed by the contracts. Contracts have multiple purposes. Primarily, they are both a legal and financial tool that protects both buyer and seller from spot price volatility. Additionally, contracts provide a predictable revenue stream that can be used as collateral for long-term financing of new projects and strengthen the commercial feasibility of existing power plants in markets with significant participation of variable RE or of energy sources with low marginal cost. Finally, contracts allocate risks, define liabilities, and offer guidance to act in cases of unanticipated contingencies.

One relevant risk for the procurement of power plants still in the project stage is the conciliation of the construction time of the power plant connection works and the deep connection works needed in the transmission grid. The combination of pre-emptive transmission corridor expansion, preliminary project elimination by connection point and a reallocation of risk to generators for transmission delays is hoped to have sufficiently decreased the likelihood of this risk being realised.

7. Conclusion and policy implications

Brazil is one of the pioneers in the systematic use of energy auctions as an instrument to support public policies. The Brazilian government structured its current auction programme in 2004 with the objectives of achieving energy security, improving the efficiency of electricity contracting for captive consumers, and promoting diversification of energy supply (especially for RES).

The auction programme achieved all initial objectives and was successful in unlocking private investment in new capacity, contracting 105 GW from greenfield power plants, and increasing the installed capacity of 2004 (89 GW) by 118 per cent. More significantly, the auction programme fostered the growth of non-conventional renewable sources in Brazil at increasingly competitive prices. The competitiveness of wind and solar technologies is partly explained by the worldwide reduction in

the equipment costs, due to technological advances and economies of scale, but also reflects the development of local investment capacity and the design of the auction programme.

Regarding the implementation process, the responsibility of each institution that takes part in the auction programme is clearly defined under a comprehensive regulatory framework and understood by all stakeholders. Additionally, the Brazilian programme has benefited from the performance of adequately funded institutions provided with qualified technical personnel. In fact, since 2004, the auctions have performed following the same rules and structure, with only minor adjustments in contractual clauses and in the bidding mechanism (mostly to deal with transmission connection delays).

In terms of auction design, the simplicity of winner selection criteria and the use of different stages for the qualification of projects and preferred bidders reduces the time consumed in the tender and mitigates the risk of litigation during the process. Moreover, because they are centralised mechanisms, the design of the auction can help to reduce some risks related to the contracting and implementation of infrastructure projects. In the case of the Brazilian experience, special attention was given to reducing the risk of off-taker payment, and to the risk of delay in the construction of projects.

The auction also seeks to mitigate construction risk by using a prequalification process, and by requiring surety and performance bonds.

Nevertheless, auctions are not able to overcome competitive market structure problems or mitigate all risks associated with the implementation of infrastructure projects. Still, it is important that contracts establish the obligations and liabilities of generators and buyers. In the Brazilian approach, most of the risks related to the environmental licensing, site selection and connection with the grid are allocated to generators, assuming that even when they are not able to manage the risk, they are better able to evaluate and price the risk.

CRedit authorship contribution statement

Maurício T. Tolmasquim: Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Tiago de Barros Correia:** Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Natália Addas Porto:** Formal analysis, Data curation, Visualization, Writing – review & editing. **Wikus Kruger:** Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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